

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 8 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 8

Subject Code & Name: BTCVSS801E - STRUCTURAL DYNAMICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
1. What is the significance of damping in controlling the vibrations of structures according to IS 1893 Indian standard code of practice on earthquake resistant design of structures?
(A) The presence of damping, however, limits the amplitudes at resonance, showing the importance of damping in controlling the vibrations of structures
(B) Damping in a structural system generally assumed to be of viscous type for mathematical convenience
(C) Damping is a measure of energy dissipation in vibration system
(D) Damping in a structural system generally assumed to be of Rigid type for mathematical convenience
 2. The natural frequency of an undamped system depends on:
(A) Damping only
(B) Mass and stiffness
(C) Force and time
(D) Displacement only
 3. Dynamic equilibrium in structural systems is represented by which type of equation?
(A) Algebraic equation
(B) Integral equation
(C) Differential equation
(D) Polynomial equation
 4. Which numerical method is commonly used for solving dynamic equilibrium equations?
(A) Finite difference method
(B) Fourier method
(C) Central difference method
(D) Euler's method

- 5 In undamped free vibration, the system's total energy remains:
 - (A) Constant
 - (B) Increasing
 - (C) Decreasing
 - (D) Zero
- 6 Response due to dynamic load is,
 - (A) Displacement
 - (B) Velocity
 - (C) Acceleration
 - (D) Displacement, Velocity and Acceleration
- 7 A Single Degree of Freedom (SDOF) system has motion described by:
 - (A) One independent coordinate
 - (B) Two independent coordinates
 - (C) Three equations of motion
 - (D) No coordinate
- 8 What will be the logarithmic decrement, if damping factor is 0.33
 - (A) 2.19
 - (B) 2.06
 - (C) 2.00
 - (D) 2.11
- 9 If, $m = 10 \text{ kg}$, $c = 10 \text{ Ns/m}$ and $k = 4 \text{ N/m}$, the system is
 - (A) Overdamped
 - (B) Critically damped
 - (C) Underdamped
 - (D) Viscous damped
- 10 Critical damping occurs when the damping ratio ζ is:
 - (A) 0
 - (B) Less than 1
 - (C) 1
 - (D) Greater 1
- 11 Newmark's method is used for:
 - (A) Calculating static deflection
 - (B) Solving nonlinear systems
 - (C) Time integration of dynamic equations
 - (D) Estimating damping ratios
- 12 The half-power bandwidth method is used to determine:
 - (A) Resonant amplitude
 - (B) Resonant frequency
 - (C) Natural period
 - (D) Damping ratio

Q2. Solve the following.

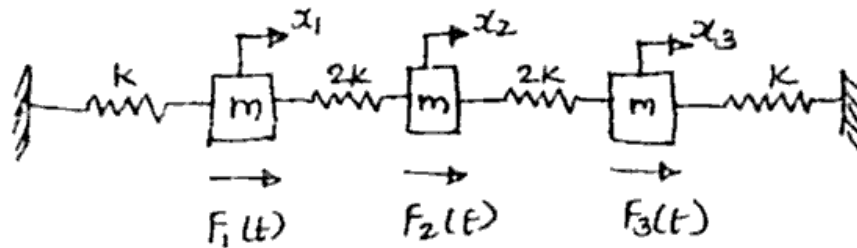
- A) State and Explain SDoF and MDoF with neat sketch. 6
- B) Draw and explain an analytical model for one degree freedom system? 6
Also draw free body diagram of it?

Q3. Solve the following.

- A) Explain undamped free vibration and derive an equation of motion for it. 6
- B) Discuss Newmark's method, including its stability and accuracy considerations. 6

Q4. Solve Any Two of the following.

- A) Explain Response Spectra and its applications. 6
- B) Three spring mass system is shown in the following figure. All the masses are subjected to dynamic forces. Derive the equation of motion in terms of displacements of the masses along the axis of the springs. 6



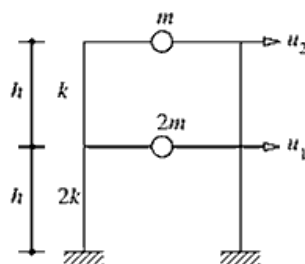
- C) A free Vibration test is conducted on an empty elevated water tank. A Cable is attached to the tank applies a lateral force of 75 kN and pulls the tank horizontally by 5 cm. The cable is suddenly cut, and the resulting free vibration is recorded. At the end of 4 complete cycles, the time is 2 second and amplitude is 2.5 cm. Take acceleration due to gravity as 10 m/second square. Calculate a) Damping ratio, b) Natural Period of undamped vibration. 6

Q5. Solve Any Two of the following.

- A) Discuss the role of structural damping in the behaviour of flexible structures subjected to dynamic loads. Provide examples. 6
- B) Discuss the challenges and advantages of analysing MDOF systems. 6
- C) A Delicate instrument weighing 0.45 kN is to be mounted on a rubber pad to the floor of a test laboratory where the Vertical acceleration 0.1 g at a frequency of 10 Hz. It has been determined experimentally that the ratio of the stiffness K to the damping coefficient $C = 100(1/\text{sec})$ for the type of rubber pad material used in the isolation. If the acceleration transmitted to the instrument is 0.01 g for 10% of damping. Calculate the natural frequency and stiffness of the system? 6

Q6. Solve Any Two of the following.

- A) Explain the concept of seismic isolation in the context of structural dynamics. How does it work, and what are its benefits in earthquake-prone regions. 6
- B) How does the addition of damping influence the mode shapes and natural frequencies of a vibrating structure? Provide a theoretical explanation. 6
- C) Determine the natural frequencies and modes of the system shown in the following Figure. 6



*** End ***